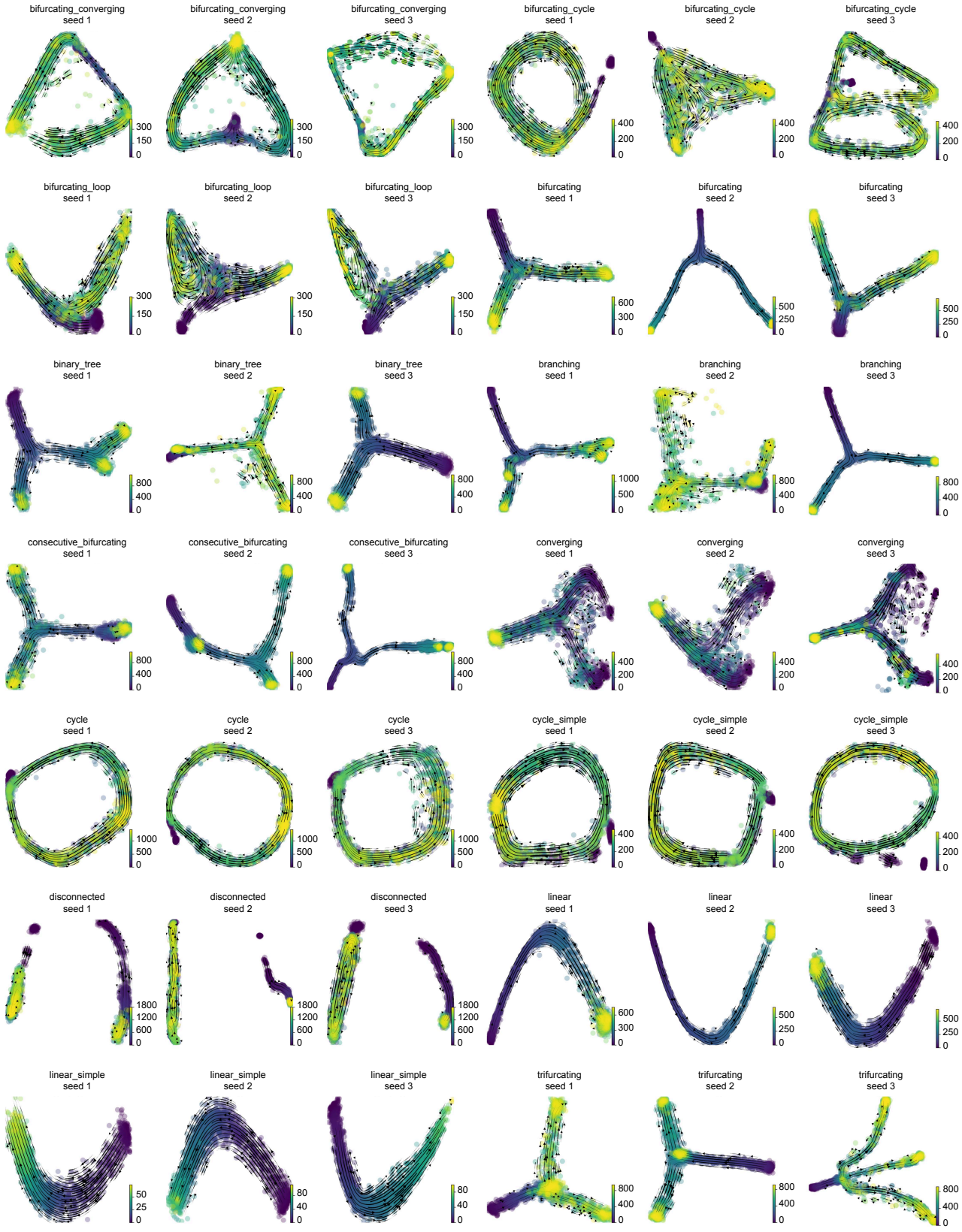
a

	Statistical Models	Deep Learning Models
Standard Input (u + s)	velocyto, scVelo, UniTVelo, TopicVelo, Cell2fate, SDEvelo	VeloAE, VeloVAE, DeepVelo (VAE-based), cellDancer, veloVI, LatentVelo, DeepVelo (GCN-based), biVI, DeepKINET
Enhanced Input (Multi-modal / Spatial / Prior information)	Dynamo, MultiVelo, PhyloVelo, TFvelo, STT	TopoVelo

b



Extended Data Fig. 1 | RNA velocity method taxonomy and dyngen simulation datasets. a, RNA velocity methods grouped by modeling strategy and input information. Categories summarize principal strategies and can overlap. Method coverage varies by evaluation task. **b,** Overview of 42 dyngen simulation datasets, covering 14 topologies with three seed datasets per topology. Velocity streams show simulated directions, with colors indicating simulated time. Dataset names and experimental design are documented in Table S2.